CORE PURE / DIFFERENTIAL EQUATIONS

Integrating Factor Method

Mark scheme 15 questions · 110 marks

Questions

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	Total	**110**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$e^{\int \frac{2x}{x^2+1}\,\mathrm{d}x} = e^{\ln(x^2+1)}$	**M1**	Forms $e^{\int P\,\mathrm{d}x}$ with the given P and integrates to the form $e^{k\ln(x^2+1)}$.
	$= x^2+1$	**A1**	
	$(x^2+1)y = \int 6x(x^2+1)\,\mathrm{d}x$ $= \ldots$	**dM1**	Multiplies through by their integrating factor, LHS of the form (their factor)$\times y$, and integrates the RHS with at least one power increased by 1.
	$y = \dfrac{\frac{3}{2}x^4+3x^2+c}{x^2+1}$	**A1**	oe, in the form $y = \ldots$, with a constant of integration.
		[4]	
1(b)	$\frac{7}{2} = \dfrac{c}{1} \Rightarrow c = \ldots\ [= \frac{7}{2}]$	**M1**	Substitutes $x = 0$, $y = \frac{7}{2}$ into their general solution containing a constant and finds the constant.
	$4(x^2+1) = \frac{3}{2}x^4 + 3x^2 + \text{“}\frac{7}{2}\text{”}$ $\Rightarrow 3x^4 - 2x^2 - 1 = 0 \Rightarrow x^2 = \ldots$	**M1**	Sets their particular solution equal to 4 and solves the resulting three-term quadratic in x^2 by any method, calculator allowed, reaching $x^2 = \ldots$
	$x = 1$	**A1**	Only $x = 1$ given as the answer.
		[3]	
Total for Question 1		**7**	

Question	Scheme	Marks	Notes
2(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2}{1+x}y = (1+x)^3$ $e^{\int -\frac{2}{1+x}\,\mathrm{d}x} = \ldots$	**M1**	Divides by $1+x$ and attempts $e^{\int P\,\mathrm{d}x}$ with their P, reaching the form $e^{k\ln(1+x)}$ or $(1+x)^k$. Condone sign slips.
	$e^{-2\ln(1+x)} = (1+x)^{-2}$	**A1**	oe, such as $\frac{1}{(1+x)^2}$; not left as $e^{-2\ln(1+x)}$.
	$\frac{\mathrm{d}}{\mathrm{d}x}\left(y(1+x)^{-2}\right) = 1+x$ $\Rightarrow y(1+x)^{-2} = \int(1+x)\,\mathrm{d}x = \ldots$	**dM1**	Multiplies their standard form by their integrating factor and integrates: LHS as $y\times$ their integrating factor, at least one RHS term with power increased by 1.
	$y = (1+x)^2\left(\frac{x^2}{2} + x + c\right)$	**A1**	Any equivalent form with an arbitrary constant, such as $y = \frac{1}{2}(1+x)^4 + c(1+x)^2$. Must be $y = \ldots$.
		[4]	
2(b)	$-1 = (1+0)^2(0+0+c) \Rightarrow c = \ldots$ $[= -1]$	**M1**	Substitutes $x=0$, $y=-1$ into their general solution with an arbitrary constant and finds the constant.
	$(1+x)^2\left(\frac{x^2}{2} + x - 1\right) = 0$ $\Rightarrow x^2 + 2x - 2 = 0 \Rightarrow x = \ldots$	**M1**	Sets their particular solution to 0 and solves the remaining quadratic factor by any means, calculator allowed, to $x = \ldots$.
	$x = \sqrt{3} - 1$	**A1**	Exact, with $-1-\sqrt{3}$ rejected or not given.
		[3]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1+x}y = \frac{\ln(1+x)}{1+x}$ $e^{\int \frac{1}{1+x}\,\mathrm{d}x} = e^{\ln(1+x)}$	**M1**	Divides by $1+x$ and forms $e^{\int P\,\mathrm{d}x}$ with $P = \frac{k}{1+x}$, or recognises the LHS as $\frac{\mathrm{d}}{\mathrm{d}x}\left((1+x)y\right)$.
	$= 1 + x$	**A1**	Or $\frac{\mathrm{d}}{\mathrm{d}x}\left((1+x)y\right) = \ln(1+x)$ stated. Accept $\lvert 1+x\rvert$.
	$(1+x)y = \int \ln(1+x)\,\mathrm{d}x$ $= x\ln(1+x) - \int \frac{x}{1+x}\,\mathrm{d}x$	**dM1**	Multiplies through by their integrating factor and integrates $\ln(1+x)$ by parts to the form $\alpha x\ln(1+x) - \int g(x)\,\mathrm{d}x$, or uses $\int \ln u\,\mathrm{d}u = u\ln u - u$ with $u = 1+x$.
	$= (1+x)\ln(1+x) - x\ (+c)$	**A1**	oe, such as $(1+x)\ln(1+x) - (1+x)$.
	$y = \ln(1+x) + \frac{c-x}{1+x}$	**A1**	cso. Any equivalent form $y = \ldots$ with the arbitrary constant divided by $1+x$. Accept $\ln\lvert 1+x\rvert$.
		[5]	
3(b)	$-1 = \ln 1 + \frac{c-0}{1+0} \Rightarrow c = -1$ $0 = \ln(1+x) - 1 \Rightarrow x = \ldots$	**M1**	Substitutes $x = 0$, $y = -1$ into their general solution containing an arbitrary constant, then sets $y = 0$ and solves for x.
	$x = e - 1$	**A1**	Or awrt 1.72.
		[2]	
	Total for Question 3	**7**	

Question	Scheme	Marks	Notes
4(a)	$\frac{\mathrm{d}}{\mathrm{d}x}\ln(\sec x+\tan x)=\frac{\sec x\tan x+\sec^2 x}{\sec x+\tan x}$	**M1**	Obtains a quotient of the form $\frac{f'(x)}{\sec x+\tan x}$ with an attempt at $f'(x)$. Condone one slip in the derivatives of $\sec x$ or $\tan x$.
	$=\frac{\sec x(\tan x+\sec x)}{\sec x+\tan x}=\sec x$	**A1 AG**	Correct derivative simplified without error to $\sec x$, by factorising or otherwise. A0 for a jump from the quotient to $\sec x$ with no simplification shown.
		[2]	
4(b)	$e^{\int -2\tan x\,\mathrm{d}x}=e^{-2\ln\sec x}$	**M1**	Attempts $e^{\int -2\tan x\,\mathrm{d}x}$, integrating $\tan x$ to $\pm\ln\sec x$ or $\pm\ln\cos x$. Condone sign slips and a missing modulus.
	$\cos^2 x$	**A1**	oe such as $\frac{1}{\sec^2 x}$; not left in exponential form.
	$y\cos^2 x=\int\sec x\,\mathrm{d}x$ $=\ln(\sec x+\tan x)+c$	**dM1**	Writes $y\times$ their integrating factor $=\int$ their integrating factor $\times\sec^3 x\,\mathrm{d}x$ and integrates, using part (a) where the integrand is $\sec x$.
	$y=\sec^2 x\big(\ln(\sec x+\tan x)+c\big)$	**A1**	$y=\ldots$ oe, with the constant also multiplied by $\sec^2 x$. Condone a modulus.
		[4]	
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5	$e^{\int -\tan x \, \mathrm{d}x} = e^{-\ln(\sec x)}$	**M1**	Integrates $\pm \tan x$ inside the exponential to the form $\pm \ln(\sec x)$ or $\pm \ln(\cos x)$. Condone sign slips and a missing modulus.
	$= \cos x$	**A1**	oe, such as $\frac{1}{\sec x}$.
	$y \cos x = \int \frac{\cos x}{\sqrt{1+\sin x}} \, \mathrm{d}x$ $= \int u^{-1/2} \, \mathrm{d}u = \ldots$	**dM1**	Writes y times their integrating factor equal to the integral of Q times it, and integrates to the form $k\sqrt{1+\sin x}$, by substitution or inspection.
	$= 2\sqrt{1+\sin x}$	**A1**	
	$y = \frac{2\sqrt{1+\sin x}+c}{\cos x}$	**A1**	oe, such as $y \cos x = 2\sqrt{1+\sin x} + c$; constant added before dividing. y need not be the subject.
	Total for Question 5	**5**	

Question	Scheme	Marks	Notes
6(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x^2}y = \frac{1}{x^2}$	**B1**	
	$e^{\int x^{-2}\,\mathrm{d}x} = e^{\cdots}$	**M1**	Uses $e^{\int P\,\mathrm{d}x}$ with their P from a form $\frac{\mathrm{d}y}{\mathrm{d}x} + Py = Q$, raising the power by 1. Not for quoting the formula alone.
	$\int x^{-2}\,\mathrm{d}x = -x^{-1} \Rightarrow e^{-1/x}$	**A1 AG**	Correct integral from a correct standard form, leading without error to $e^{-1/x}$.
		[3]	
6(b)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(ye^{-1/x}\right) = \frac{e^{-1/x}}{x^2}$	**M1**	Multiplies their standard form by $e^{-1/x}$ and writes LHS as $\frac{\mathrm{d}}{\mathrm{d}x}(ye^{-1/x})$. Condone slips on RHS.
	$ye^{-1/x} = \int \frac{e^{-1/x}}{x^2}\,\mathrm{d}x = \ldots$	**dM1**	Integrates both sides, RHS to the form $ke^{-1/x}$, by inspection or substitution. The constant need not appear.
	$ye^{-1/x} = e^{-1/x} + c \Rightarrow y = 1 + ce^{1/x}$	**A1**	oe, including the unrearranged form. Needs the constant.
	$3 = 1 + ce \Rightarrow c = \ldots$	**M1**	Substitutes $x = 1$, $y = 3$ into their general solution with an arbitrary constant and solves for the constant.
	$y = 1 + 2e^{1/x-1}$	**A1**	oe, such as $y = 1 + \frac{2}{e}e^{1/x}$, or with the coefficient awrt 0.736.
		[5]	
Total for Question 6		**8**	

Question	Scheme	Marks	Notes
7	$\frac{\mathrm{d}y}{\mathrm{d}x} - y\coth x = \cosh x$ $e^{-\int \coth x\,\mathrm{d}x} = e^{-\ln(\sinh x)}$	**M1**	Divides by $\sinh x$ and attempts $e^{\int P\,\mathrm{d}x}$ with $P = \pm\coth x$, integrating to the form $\pm\ln(\sinh x)$. Condone a sign slip.
	$= \frac{1}{\sinh x}$	**A1**	oe, such as $\operatorname{cosech} x$.
	$\frac{y}{\sinh x} = \int \coth x\,\mathrm{d}x$	**dM1**	Writes $y\times$ their integrating factor $= \int$ (their integrating factor $\times \cosh x$) $\mathrm{d}x$ and attempts the integral.
	$\frac{y}{\sinh x} = \ln(\sinh x) + c$	**A1**	Any equivalent form, with the constant.
	$\sinh\big(\ln(1+\sqrt{2})\big) = 1$ $\Rightarrow 2 = 1(\ln 1 + c) \Rightarrow c = \ldots$	**M1**	Substitutes $x = \ln(1+\sqrt{2})$, $y = 2$ into their general solution with a constant, evaluating $\sinh\big(\ln(1+\sqrt{2})\big)$ by any means, to find c.
	$y = \sinh x\big(\ln(\sinh x) + 2\big)$	**A1**	Any equivalent exact form, such as $y = \sinh x \ln(\sinh x) + 2\sinh x$.
	Total for Question 7	**6**	

Question	Scheme	Marks	Notes
8	$\frac{\mathrm{d}y}{\mathrm{d}x} - 2\coth x\, y = \sinh x \cosh x$ $e^{\int -2\coth x\,\mathrm{d}x} = e^{-2\ln(\sinh x)}$	**M1**	Divides by $\sinh x$ and attempts $e^{\int P\,\mathrm{d}x}$ with $P = k\coth x$, integrating to a multiple of $\ln(\sinh x)$. Condone sign and coefficient slips.
	$= \frac{1}{\sinh^2 x}$	**A1**	oe, such as $\mathrm{cosech}^2 x$.
	$\frac{y}{\sinh^2 x} = \int \coth x\,\mathrm{d}x$	**dM1**	Multiplies through by their integrating factor, writes the LHS as $y\times$ their integrating factor, and attempts to integrate the RHS.
	$\frac{y}{\sinh^2 x} = \ln(\sinh x) + c$	**A1**	oe, with the constant. Condone $\ln\lvert\sinh x\rvert$.
	$\sinh\big(\ln(1+\sqrt{2})\big) = 1$ $5 = 1^2(\ln 1 + c) \Rightarrow c = \ldots$	**M1**	Substitutes $y = 5$ and $x = \ln(1+\sqrt{2})$ into their general solution containing a constant and solves for it.
	$y = \sinh^2 x\big(\ln(\sinh x) + 5\big)$	**A1**	Any equivalent exact form.
	Total for Question 8	**6**	

Question	Scheme	Marks	Notes
9(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} - \coth x\, y = \cosh^3 x$	**B1**	
	$e^{\int -\coth x \,\mathrm{d}x} = e^{-\ln(\sinh x)}$	**M1**	Integrates their P, of the form $k\coth x$, to $k\ln(\sinh x)$ and forms $e^{\int P\,\mathrm{d}x}$. Condone a sign slip.
	$\frac{1}{\sinh x}$	**A1**	oe, such as $\operatorname{cosech} x$. Condone $\lvert\sinh x\rvert$.
		[3]	
9(b)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \frac{\cosh^3 x}{\sinh x}$	**M1**	Multiplies by their integrating factor μ, a function of x, to reach $\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \mu Q$, or writes $\mu y = \int \mu Q\,\mathrm{d}x$.
	$\frac{y}{\sinh x} = \int \frac{\cosh x(1+\sinh^2 x)}{\sinh x}\,\mathrm{d}x$ $= \int \left(\frac{1}{u} + u\right)\mathrm{d}u = \ldots$	**dM1**	Uses $\cosh^2 x = 1 + \sinh^2 x$ and integrates the right side to the form $p\ln(\sinh x) + q\sinh^2 x$, $q\cosh^2 x$ or $q\cosh 2x$, by any valid method. Condone coefficient slips.
	$\ln(\sinh x) + \frac{1}{2}\sinh^2 x$	**A1**	oe, such as $\frac{1}{4}\cosh 2x$ or $\frac{1}{2}\cosh^2 x$ for $\frac{1}{2}\sinh^2 x$.
	$y = \sinh x \ln(\sinh x) + \frac{1}{2}\sinh^3 x + c\sinh x$	**A1**	Needs the constant; y need not be the subject.
	$\sinh(\ln(1+\sqrt{2})) = 1$ $\Rightarrow 2 = 0 + \frac{1}{2} + c \Rightarrow c = \ldots$	**M1**	Substitutes $y = 2$ and $\sinh x = 1$, from the exponential definition or a calculator, into a general solution with a constant, to find it.
	$y = \sinh x \ln(\sinh x) + \frac{1}{2}\sinh^3 x + \frac{3}{2}\sinh x$	**A1**	Any equivalent exact form for y; c depends on the form of their integral.
		[6]	
	Total for Question 9	**9**	

Question	Scheme	Marks	Notes
10(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} + y\tan x = x\sec x$	**B1**	
	$e^{\int \tan x\,\mathrm{d}x} = e^{\ln(\sec x)}$	**M1**	Forms $e^{\int P\,\mathrm{d}x}$ with P the coefficient of y in their standard form, integrating to a log form such as $\pm\ln(\cos x)$.
	$= \sec x$	**A1 AG**	cso from a correct integral to $\sec x$. Condone omitted modulus signs.
		[3]	
10(b)	$\sec x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sec x\tan x = x\sec^2 x$ $\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sec x) = x\sec^2 x$	**M1**	Multiplies their standard form by $\sec x$ and writes LHS as $\frac{\mathrm{d}}{\mathrm{d}x}(y\sec x)$; implied by $y\sec x = \int x\sec^2 x\,\mathrm{d}x$.
	$y\sec x = x\tan x - \int \tan x\,\mathrm{d}x$	**dM1**	Integrates by parts with $u = x$ to the form $x\tan x \pm \int \tan x\,\mathrm{d}x$. Condone sign slips.
	$\int x\sec^2 x\,\mathrm{d}x = x\tan x + \ln(\cos x)$	**A1**	oe, such as $x\tan x - \ln(\sec x)$; condone omitted modulus.
	$y = x\sin x + \cos x\ln(\cos x) + c\cos x$	**A1**	oe, including $y\sec x = x\tan x + \ln(\cos x) + c$. Constant must be present.
	$\sqrt{2} = \frac{\pi}{4} + \ln\frac{\sqrt{2}}{2} + c \Rightarrow c = \ldots$	**M1**	Substitutes $x = \frac{\pi}{4}$, $y = 1$ into their general solution containing a constant and solves for it.
	$y = x\sin x + \cos x\left[\ln(\cos x) + \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2\right]$	**A1**	oe, including $y\sec x = \ldots$; constant exact or awrt 0.975.
		[6]	
	Total for Question 10	**9**	

Question	Scheme	Marks	Notes
11	$e^{\int \cot x \, \mathrm{d}x} = e^{\ln(\sin x)}$	**M1**	Uses $e^{\int \cot x \, \mathrm{d}x}$ with $\cot x$ integrated to the form $k\ln(\sin x)$. Condone a missing modulus.
	$\sin x$	**A1**	Accept $\lvert \sin x \rvert$.
	$\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) = \frac{\cos x}{\sin x(1+[\ln(\sin x)]^2)}$ $u = \ln(\sin x) \Rightarrow y\sin x = \int \frac{1}{1+u^2}\,\mathrm{d}u$	**dM1**	Multiplies through by their integrating factor and integrates both sides, the RHS by a substitution such as $u = \ln(\sin x)$ or by inspection. Condone slips in the substitution.
	$y\sin x = \arctan(\ln(\sin x))$	**A1**	Correct integral; the constant need not appear yet.
	$y = \frac{\arctan(\ln(\sin x))+c}{\sin x}$	**A1**	oe, such as $y\sin x = \arctan(\ln(\sin x)) + c$. Needs the constant.
	Total for Question 11	**5**	

Question	Scheme	Marks	Notes
12(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x+1}{x(x-1)}y = x-1$	**B1**	RHS need not be simplified.
	$\frac{x+1}{x(x-1)} = \frac{A}{x} + \frac{B}{x-1}$ $\Rightarrow x+1 = A(x-1) + Bx \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Uses partial fractions of the form $\frac{A}{x} + \frac{B}{x-1}$ and finds both constants by substitution or equating coefficients. Condone arithmetic slips.
	$\frac{x+1}{x(x-1)} = -\frac{1}{x} + \frac{2}{x-1}$	**A1**	
	$e^{\int\left(-\frac{1}{x}+\frac{2}{x-1}\right)\mathrm{d}x} = e^{-\ln x + 2\ln(x-1)}$	**M1**	Forms $e^{\int P\,\mathrm{d}x}$ with their partial fractions integrated to the form $p\ln x + q\ln(x-1)$. Condone missing modulus signs.
	$= \frac{(x-1)^2}{x}$	**A1 AG**	Correct exponent leading by log laws without error to the printed result. Modulus signs not required.
		[5]	
12(b)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x-1)^2}{x}y\right) = \frac{(x-1)^3}{x}$	**M1**	Multiplies the standard form by the integrating factor, with LHS as $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x-1)^2}{x}y\right)$ and RHS their Q times the integrating factor.
	$\frac{(x-1)^2}{x}y = \int\left(x^2 - 3x + 3 - \frac{1}{x}\right)\mathrm{d}x$ $= \ldots$	**dM1**	Writes the integrand as a polynomial plus $\frac{k}{x}$, oe by substitution, and integrates, with at least one power increased by 1 or $\frac{k}{x} \to k\ln x$.
	$\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x$	**A1**	Condone $\ln\lvert x\rvert$.
	$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + c\right)$	**A1**	oe; y need not be the subject.
	$2 = \frac{2}{1^2}\left(\frac{8}{3} - 6 + 6 - \ln 2 + c\right) \Rightarrow c = \ldots$ $\left[c = \ln 2 - \frac{5}{3}\right]$	**M1**	Substitutes $x=2$, $y=2$ into their general solution containing a constant of integration and solves for c.
	$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + \ln 2 - \frac{5}{3}\right)$	**A1**	oe with y the subject, such as $\ln\frac{2}{x}$, or constant awrt -0.974.
		[6]	
	Total for Question 12	**11**	

Question	Scheme	Marks	Notes
13(a)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\cosh x+1)\right)=\frac{\sinh x}{\cosh x+1}$	**M1**	Uses the chain rule to obtain the form $\frac{k\sinh x}{\cosh x+1}$. Condone a sign slip.
	$=\frac{2\sinh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{2}\right)}{2\cosh^2\left(\frac{x}{2}\right)}$	**M1**	Writes both $\sinh x$ and $\cosh x+1$ in terms of $\frac{x}{2}$ by double-angle identities or exponential definitions. Condone coefficient slips.
	$=\tanh\left(\frac{x}{2}\right)$	**A1 AG**	Correct derivative and identities leading without error to $\tanh\left(\frac{x}{2}\right)$; combined steps accepted.
		[3]	
13(b)	$e^{\int\tanh\left(\frac{x}{2}\right)\mathrm{d}x}=e^{\ln(\cosh x+1)}=\cosh x+1$ $\frac{\mathrm{d}}{\mathrm{d}x}\left((\cosh x+1)y\right)=(\cosh x+1)\sinh x$	**M1**	Uses part (a) to obtain integrating factor $\cosh x+1$ and multiplies the RHS by it, or writes the LHS as $\frac{\mathrm{d}}{\mathrm{d}x}\left((\cosh x+1)y\right)$.
	$(\cosh x+1)y=\frac{1}{2}(\cosh x+1)^2+c$	**dM1 A1**	**dM1** Integrates both sides, LHS to $(\cosh x+1)y$ and RHS to a form such as $k(\cosh x+1)^2$, $p\cosh^2 x+q\cosh x$ or $p\cosh 2x+q\cosh x$. **A1** Correct, oe in any arrangement; needs $+c$.
	$2\times 2=\frac{1}{2}\times 2^2+c\Rightarrow c=\ldots\ [=2]$	**M1**	Substitutes $x=0$, $y=2$ into their general solution with a constant and solves for it.
	$y=\frac{\cosh x+1}{2}+\frac{2}{\cosh x+1}$	**A1**	Exact, oe such as $y=\frac{\frac{1}{4}\cosh 2x+\cosh x+\frac{11}{4}}{\cosh x+1}$; must be y in terms of x.
		[5]	
	Total for Question 13	**8**	

Question	Scheme	Marks	Notes
14(a)	$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\coth\frac{x}{2}\right)$ $= \frac{1}{\coth\frac{x}{2}} \times \left(-\frac{1}{2}\operatorname{cosech}^2\frac{x}{2}\right)$	**M1**	Applies the chain rule to reach the form $k\,\frac{\operatorname{cosech}^2(x/2)}{\coth(x/2)}$, or an equivalent via exponentials. Condone slips in k or sign.
	$= -\frac{1}{2\sinh\frac{x}{2}\cosh\frac{x}{2}} = -\frac{1}{\sinh x}$	**M1**	Writes in terms of $\sinh\frac{x}{2}$ and $\cosh\frac{x}{2}$ and uses the double-angle identity, or exponentials, to reach a function of x.
	$= -\operatorname{cosech} x$	**A1 AG**	Correct chain-rule derivative leading without error to the printed result; combined steps accepted.
		[3]	
14(b)	$\mu = e^{\int -\operatorname{cosech} x\,\mathrm{d}x} = \coth\frac{x}{2}$ $\coth\frac{x}{2}\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\coth\frac{x}{2}\,y$ $= \coth\frac{x}{2}\tanh\frac{x}{2}$	**M1**	Uses (a) to obtain the integrating factor $\coth\frac{x}{2}$ and multiplies the equation through. Condone a sign slip giving $\tanh\frac{x}{2}$.
	$y\coth\frac{x}{2} = \int \coth\frac{x}{2}\tanh\frac{x}{2}\,\mathrm{d}x = \ldots$	**dM1**	Writes $y\times$ their integrating factor $= \int$ their integrating factor $\times \tanh\frac{x}{2}\,\mathrm{d}x$ and integrates the RHS.
	$y\coth\frac{x}{2} = x + c$ $\Rightarrow y = (x+c)\tanh\frac{x}{2}$	**A1**	Either form, with the constant.
	$\tanh\frac{\ln 3}{2} = \frac{3-1}{3+1} = \frac{1}{2}$ $1 = (\ln 3 + c)\times\frac{1}{2} \Rightarrow c = \ldots$	**M1**	Substitutes $x = \ln 3$, $y = 1$ into their general solution with a constant and solves for c; a calculator value of $\tanh\frac{\ln 3}{2}$ is enough.
	$y = (x + 2 - \ln 3)\tanh\frac{x}{2}$	**A1**	Exact, oe, as y in terms of x; c alone does not score.
		[5]	
	Total for Question 14	**8**	

Question	Scheme	Marks	Notes
15(a)	$\ln(\cosh x + 1) - \ln(\sinh x)$ $\Rightarrow \frac{\sinh x}{\cosh x+1} - \frac{\cosh x}{\sinh x}$	**M1**	Differentiates using $\frac{f'}{f}$, by splitting the logarithm, reaching $\frac{\sinh x}{\cosh x+1} - \frac{\cosh x}{\sinh x}$, or by the quotient rule a single fraction $\frac{\sinh^2 x - \cosh x(\cosh x+1)}{\sinh x(\cosh x+1)}$. Condone sign slips.
	$\frac{\sinh^2 x - \cosh^2 x - \cosh x}{\sinh x(\cosh x+1)}$ $= \frac{-1-\cosh x}{\sinh x(\cosh x+1)}$	**M1**	Combines over a common denominator and uses $\cosh^2 x - \sinh^2 x = 1$ to simplify the numerator.
	$= -\frac{1}{\sinh x} = -\operatorname{cosech} x$	**A1 AG**	Correct derivative leading without error to the printed result; combined steps accepted.
		[3]	
15(b)	$\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\, y = \frac{1-\cosh x}{\sinh x}$ $\mu = e^{\int -\operatorname{cosech} x\, \mathrm{d}x} = \frac{\cosh x+1}{\sinh x}$ $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh x+1}{\sinh x}y\right)$ $= \frac{(\cosh x+1)(1-\cosh x)}{\sinh^2 x}$	**M1**	Divides by $\sinh x$, uses part (a) for the integrating factor $\frac{\cosh x+1}{\sinh x}$ and multiplies through. Condone slips in the RHS.
	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh x+1}{\sinh x}y\right) = -1$ $\Rightarrow \frac{\cosh x+1}{\sinh x}y = \ldots$	**dM1**	Simplifies the RHS using $\cosh^2 x - \sinh^2 x = 1$ and integrates both sides. Condone a sign slip.
	$\frac{\cosh x+1}{\sinh x}y = -x + c$	**A1**	Correct, with $+c$, in any equivalent form.
	$\sinh(\ln 3) = \frac{4}{3},\ \cosh(\ln 3) = \frac{5}{3}$ $\frac{\frac{8}{3}}{\frac{4}{3}} \times 2 =$ "$c - \ln 3$" $\Rightarrow c = \ldots\ [= 4 + \ln 3]$	**M1**	Substitutes $x = \ln 3$, $y = 2$ into their general solution containing c, evaluates $\sinh(\ln 3)$ and $\cosh(\ln 3)$, and solves for c. Condone arithmetic slips.
	$y = \frac{(4+\ln 3-x)\sinh x}{\cosh x+1}$	**A1**	Any exact equivalent, e.g. $y = (4 + \ln 3 - x)\tanh\frac{x}{2}$.
		[5]	
	Total for Question 15	**8**	